From Community Feedback to Measurement Redesign

Redesigning mRehab for accessible stroke rehabilitation at home

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What is mRehab?

After a stroke, arm and hand rehabilitation often moves from the clinic to the home before recovery is complete. Nearly two-thirds of survivors do not complete their prescribed home exercises. mRehab places a smartphone inside 3D-printed household objects for practicing everyday tasks.



Figure A: *A phone inside a 3D-printed mug guides the exercise and records the movement.*

We asked 22 community experts to review mRehab, then mapped their feedback to changes in the app, the props, and the software. Some changes are built; others are still planned.

Who gave feedback?

One two-hour Community Engagement Studio with the Clinical and Translational Science Institute (CTSI), August 2025: a presentation, hands-on use, and a guided discussion.

22
community expert consultants

81
coded feedback segments

Consultants were aged 21 to 70 and had a connection to stroke care or disability advocacy:

- Eight patient and disability advocates.
- Six health professionals.
- Five medical students.
- Three physical therapists.

HOW WE CODED THE FEEDBACK

An LLM made a first pass over the transcript. The fixed rubric recorded:

- Speaker role and session phase.
- Comment type and theme.

One coder reviewed every label. A segment is one comment; a consultant could contribute several.

HOW OFTEN EACH THEME CAME UP

Theme	Segments
Personalization	19
Accessibility	13
Hardware	12
Clinical integration	11
Access and equity	9
Market reach	8
Evidence	5
Audio/voice	2
AI features	2

These count comments, not people. Consultants could raise several themes; personalization was the most frequent.

From feedback to design

How we responded to each theme, and which changes still need work.

Theme	Our response	Status
Personalization	Per-user difficulty and session-history recall.	**Planned**
Accessibility	The voice instruction repeats when no movement is detected. Voice-first, captioned, and low-touch use.	**Built** **Planned**
Hardware	Bowl holder angle changed from 90° to 45°.	**Built; testing**
Clinical integration	Activity logic configurable per module. App flags non-starts, incomplete repetitions, unstable baselines, and irregular movement.	**Built**
Access and equity	Offline use, affordability, several languages, and setup help.	**Required before deployment**
Evidence	Records keep repetition counts, duration, RMS jerk, and sensor traces.	**Built**

Smaller themes, market reach, audio pacing, and AI-voice requests, are future targets.

The largest theme is still open. Personalization drew the most feedback, and none of it is built yet.

WHAT CONSULTANTS ASKED FOR

Consultants valued real tasks at home. They asked for:

- Larger text and slower, louder prompts for people with hearing or vision loss.
- Less fine touch input, with aphasia also considered in the interaction design.
- Props that keep phones in place and fit different grips and handedness.
- A viewing angle that lets people see the phone during an activity.
- Therapists choosing activities and setting the difficulty.

WHAT IS READY TO TEST?

Repeated instructions and activity logic are built. The bowl holder needs testing; personalization is planned.

Why change the filter?

The Evidence theme most directly shaped this stage: trustworthy, quantitative feedback. Hands-on demonstrations and recorded traces showed how delayed activity acknowledgment could weaken trust. These were a separate source from the 81 coded segments.

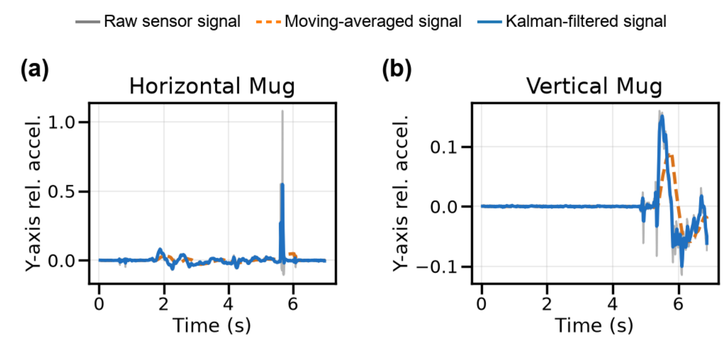


Figure B: *(a) Horizontal mug, (b) vertical mug. Kalman tracks movement sooner than the moving average; the horizontal peak still shrinks.*

What the filter changes

The phone samples its motion sensors at 50 Hz. We replaced the moving average (MA) with a scalar Kalman filter (KF) on each axis and compared both on traces from four mug activities.

0–17 ms
Kalman lag, vs. 204–221 ms for the moving average

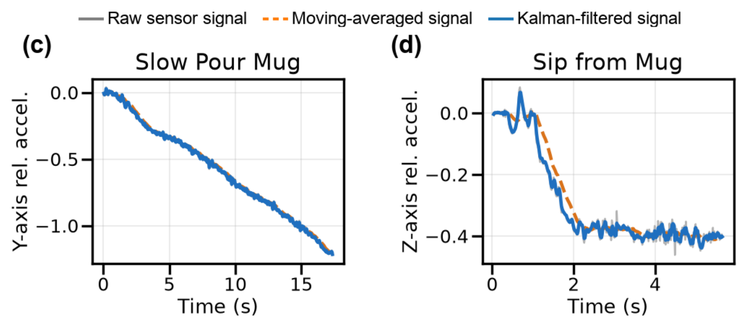


Figure C: *(c) Slow-pour mug, (d) sip from mug. The filters track the slow pour similarly; Kalman follows the sip more closely.*

Activity	MA lag	KF lag	MA loss	KF loss
Horizontal mug	204 ms	0 ms	89%	48%
Vertical mug	204 ms	17 ms	31%	3%
Slow pour mug	204 ms	0 ms	0%	0%
Sip from mug	221 ms	0 ms	10%	4%

READING THE MEASUREMENTS

- Lag measures how far the filtered signal trails the raw one.
- A lag of 0 ms means no lag was measurable in these traces.
- Peak loss measures how much the filter shrinks a peak.

Lower lag does not preserve every peak. Kalman still lost 48% of the horizontal-mug peak, so this activity needs particular attention in further testing.

REPETITIONS AND SMOOTHNESS

- The app saves a resting baseline for each axis.
- It counts a repetition when movement returns to rest.
- If movement has not started, it repeats the instruction.
- RMS jerk scores smoothness. Lower values mean smoother movement.

Old and new jerk scores are not comparable. Changing the filter changes the signal used to calculate smoothness, so the scores need separate interpretation.

Limits and next steps

This preliminary trace-level check explains why we chose Kalman filtering. It is not bench-top validation or evidence of clinical effectiveness.

The consultants gave expert views, not direct testimony from stroke survivors. We still need to test the revised system with the people who will use it at home.

- Pilot the revised bowl holder with survivors to test the changed angle during use.
- Compare repetition counts and RMS jerk against a wearable motion sensor (IMU). This will check the measurements beyond the phone traces.
- Build per-user difficulty and session-history recall, the response to the largest feedback theme.
- Add multilingual, voice-first interaction so the planned accessibility changes can be tested in use.

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